UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / - 2020

Roll No.

Paper: Computational Physics-I

Course Code: PHY-311 Part – II

Time: 2 Hrs. 45 Min. Marks: 60

ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Q.2.	Write short answers of the following Questions:	
i.	Explain with example the following terms in C++: (a) >> (b) << (c) // (d) ;	4
ii.	Write syntax with example the following in C++: (a) void main ( ) (b) bool	4
iii.	Write at least four rules for variable declaration in C++?	4
	Write C++ program code segment for the following: (a) to read and convert length from kilometers to meters by user defined function (b) to print x, y and z against f(x,y,z) for the equation $f(x,y,z) = zx^3 + xy - yz^3$ read x, y and z from the user (c) To print sum of reciprocals of factorials for numbers 1 to 5	8
Q.3.	Write C++ program to evaluate the $\int_{1}^{5}(x^3)dx$ by Trapezoidal's rule or by Weddle's (1/3) rule due to options 1 or 2 respectively (use n=6). Report error message for any other option pressed by the user.	10
Q.4.	Suppose A and B be 4x4 matrices. Write C++ program which reads in entries of A and B and prints out the entries of matrix which is (i) $C = A \bullet B$, (ii) $D = 0.4(A + B') + 3B$ and (iii) to print maximum of diagonal elements of C. (iv) maximum of the elements of matrix D. (v) print out 3rd row and 2nd column elements of matrix C	2+4+4
Q.5.	Find the roots of the equation $x^2 - 2x - 2$ using Newton Raphson method. Take initial values from the user. Write C++ program to implement the method correct to i) 3dp and ii) correct to 2dp. Also print out number of iterations in i) and ii). Write C++ program which reads in 10 random numbers into an array a and display: sum and product of the elements of a. Also print elements of a in reverse order. Execute the program iteratively for 5 times.	6+4

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Computational physics
—2020—

Question #03:-

$$I = \int_1^6 x^2 dx \qquad n=6$$

```
#include <iostream.h>
# include <conio.h>
#include <math.h>
void main()
{
    cout << "trapozoidal rule" << endl;
    float h, n, low, up, sum=0;
    cout << "enter lower limit";
    cin >> low;
    cout << "Enter upper limit";
    cin >> up;
    cout << "Enter number of
            intervals";
    cin >> n;
    h = (up-low)/n;
    cout << "Enter step size";
    cin >> h;
```


CS CamScanner

```
Sum = f(low) + f(up);
for( int i=1 ; i<=n; i++)
   {
     Sum += 2*f( low + i*h);
     Sum = Sum * h/2 ;

   }
   Cout << "I" << "\t" << sum = << endl;
     getch();
   float f ( float x)
   {
       float b;
         b = (x*x);
       return b;
         }
```

Question # 04

(i)

C = A*B

```
# include <iostream.h>
# include <conio.h>
# include <math.h>
 void main()
```

```
{
  int A[4][4], B[4][4], C[4][4];
  int i, j, k;
  cout << "Enter elements of
          matrix A";
  for (i=0; i<4; i++)
   {
     for (j=0; j<4; j++)
     {
        cin >> A[i][j];
     }
   }
  cout << "Enter element of
          matrix B";
  for (i=0; i<4; i++)
   {
     for (j=0; j<4; j++)
      {
        cin >> B[i][j];
      }
   }
  // Matrix Multiplication
```

```
for(i=0;i<4 ;i++)
 {
  for(j=0 ; j<4 ;j++)
   {
    C[i][j] =0;

   for (k=0; k<4; k++)
  {
   C[i][j] += A[i][k]*B[k][j] ;
    }
     }
      }
cout<< "Product matrix C:";
for (i=0; i<4; i++)
 {
  for (j=0; j<4;j++)
   {
    cout<< C[i][j] <<endl;
    }
   }
getch();
    }
```

(ii) $D = 0.4(A+B^t)+3B$

```
#include <iostream.h>
#include <conio.h>
void main()
 {
   int A[4][4], B[4][4], D[4][4];
   int i, j, transpose B[4][4];
  cout<< "Enter element of
     matrix A";
  for(i=0 ; i<4 ; i++)
 { for(j=0 ; j<4; j++)
     {
        cin>> A[i][j];
        }
       }
   cout<<"Enter element of matrix B";
   for(i=0; i<4; i++)
 {
  for(j=0; j<4; j++)
   {  cin>> B[i][j];
      }
        }
```

```
// Calculate transponse of B
for(i=0; i<4; i++)
{ for (j=0; j<4; j++)
  {
     transpose B[j][i] = B[i][j];
     }
   }
// Calculate D = 0.4(A+transpose B)+3B
 for (i=0; i<4; i++)
  {
   for (j=0; j<4; j++)
   {
  D[i][j] = 0.4*(A[i][j]+transpose B[i][j])
            + 3 * B[i][j];
       }
     }
 cout << "Matrix D=";
for(i=0; i<4; i++)
{
 for (j=0; j<4; j++)
  {
   cout << D[i][j] << endl;
       }
         }
```

```
getch();
}
```

(iii) Program to print Maximum of diagonal elements of 4x4 matric C:

```
#include<iostream.h>
#include <conio.h>
void main( )
{
  int C[4][4];
  int i, Max diagonal=0;
  cout<<"Enter the elements
  of matrix C";
  for(i=0; i<4; i++)
  {
  for (j=0; j<4; j++)
   {
     cin>> C[i][j];
   }
  }
```

// Find Maximum Digonal elements.

```
for (i=0; i<4; i++)
 {
  for(j=0; j<4; j++)
   {
    if (C[i][j]>max Digonal)
    {
     max digonal = C[i][j];
     }
   }
 cout << "Maximum of Digonal
        elements:" <<maxDigonal <<
        endl;
      }
   getch();
      }
```

(iv) Program to print 3rd row and 2nd column elements of 4x4 matrix C:

```
#include <iostream.h>
#include<conio.h>
void main()
{
  int i, C[4][4];
  cout<<"Enter element of Matrix C:";
  for(i=0; i<4; i++)
  {
  for(j=0; j<4; j++)
  {
     cin>> C[i][j];
     }
  }
  Cout << "3rd row elements:";
```

```
for(i=0; i<4; i++)
{
//Note: Array indices start from
0, so 3rd row is at index 2.
cout << C[2][i] << endl;
}
cout << "2nd column elements:";
for (i=0; i<4; i++)
{
cout << C[i][1] << endl;
// array indices start from
0, so 2nd column is at index 1
}
getch();
}
```

Question # 05

(a)

```
#include <iostream.h>
#include <conio.h>
Void main()
 {
   double x, inatial, pocision;
   int iterations=0;
   cout << "Enter the precision";
   cin>> precision;
     x = inatial;
  do
  {
   double xNext = x - f(x)/fDerivative(x)
     iteration++;
      x = xNext;
    }
 while ( precision == 1: fabs f(x) >pow(10,-3):
```

```
fabs(f(x)) > pow(10,-2));
cout << "The root is:" << x << endl;
cout << "No. of iterations:" << iteration << endl;
getch();
}
```

```
float f(float x)
{ b = x*x - 2*x - 2;
return b;
}
```

5(b):

```
#include <iostream.h>
#include <conio.h>
void main()
{ int a[10];
int sum=0, product=1;
for(int i=0; i<5; i++)
{
cout << "Iteration" << endl;
for(j=0; j<10; j++)
{
a[j] = rand() % 100;
```

```
// generate random numbers b/w
   0 and 99
        sum+ = a[j];
        product * = a[j];
        }
    cout << "sum:" << sum << endl;
    cout << "product:" << product << endl;
    cout << "Element in reverse order:";
    for( int j=9 ; j>=0; j--)
    {
        cout << a[j] << endl;
      }
        sum=0;
        product= 1;
        }
      getch();
        }
```

Short Question # 01

(i) << → Insertion.

The << operator in C++ is used for insertion or extraction, depending on the content.

```
void main()
  {
    int a=5;
    char c='A';
    string str = "Hello";
    cout << "Ther value of a is" << a <<e
    cout << "The value of c is" << c <<e
    cout << "The string is" << str << en
    getch();
  }
```

(ii) >> → Extraction.

```
void main()
{ int a;
  char c;
  cout << "Enter an integer:";
  cin >> a;
  cout << "Enter an character:"
  cin >> c;
  cout << "Enter a string";
  cin >> str;
  cout << "You entered:" << \t << a
     << \t << c << endl;
  getch();
}
```

(iii) //

The symbol (//) is used for single line comment in C++.

```
void main()
{
```

```
// declare and intialize a variable
    int a = 5;
// declare and intialize another
variable
    int b= 10;
// add a and b and store in a
    a = a+b;
// Print result.
cout << "The result is" << a << endl;
    getch();
    }
```

(iv) ;

The ; is use to terminate statement in C++.

```
e.g:{ int a = 5, b= 7;
  intsum=0;
  sum = a+b;
  cout << "sum" << sum << endl;
  getch(); }
```

(ii)

1) void main ()

Syntax of void main() in C++.

```
void main()
{
  // code to be executed
}
```

e.g:

```
void main()
{
  cout << "Hello, World!";
  getch();
}
```

2) bool

Syntax

```
bool variable_name;
```

e.g:

```
void main()
```

```
{
 bool isAdmin = true;
  if (isAdmin)
   {
   cout << "You are an admin";
    }
 else
   { cout << "You are not an admin";
    }
    getch();
      }
```

(iii)

(a) Rules for variable decleration:

1) Variable names must start with a letter or underscore(_)

e.g int myvariable; (valid)

int 2myvariable; (Invalid)

2) Variable name must be unique with in their scope

e.g

```
void main ( )
{
    int x;  // declare x
    int x;  // error: x is already
            declared
    getch ( );
}
```

3) Variable declarations must include a data type. When declaring a variable, you must specify its data type

e.g

```
int myVariable; (Valid)
myVariable; (Invalid)
```

4) Variables must be declared before they are used. This means you cannot use a variable before it has been declared.

e.g

```
void main()
{
// error: myVariable no declared
   intvariable = 5;
// declare myvariable
   int myVariable;
}
```

iv → (C)

```
#include <iostream.h>
#include <conio.h>
void main()
```

```
{  double fact = 1.0;
for (int i=1 ; i<=n ; i++)
  {
    fact *= i;
  }
for (int i=1 ; i<=5 ; i++)
{
 sum += 1.0 / fact(i);
  }
cout << "sum of reciprocal of
factorial:" << sum << endl;
   getch();
    }
```

(b)

```
void main()
  {  double x, y, z;
     cout << "Enter x:" ;
```

```
cin>> x;
cout << "Enter y:" ;
cin>> y;
cout << "Enter z:";
cin>> z;
    double f = z*x*x + x*y - y*z*z;
    cout << "x= " << x << "y= " << y <<
      "z= " << z << endl;
      getch();
   }
```

(a)

```
#include <iostream.h>
#include <conio.h>
#include <math.h>
    float f(float a)
    {
       m = Kilometers / 1000;
```

```
return m;
    }
Void main()
   { double Kilometers;
  cout<<"Enter the lenght in Km";
  cin>> km;
   m = km/1000;
   cout<< Kilometers << Kilometers
    is equal to:" <<"meters" <<endl;
    getch();
     }
```